SIMATS ENGINEERING
ASSESSMENT TOOL 1

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COURSE NAME: CSA1603 **COURSE CODE:** Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Siva Dakshitha K / **192425395***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Siva Dakshitha K
Signature of the student:

Activity - 4 25/07/26

CO2 - ALT(1) - Analytical Problems

(1) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(a) What is the mean of the data? What is the median?

(b) What is the mode of the data? comment on the data's modality (i.e., bimodal, trimodal, etc..)

(c) What is the midrange of the data?

(d) Can you Find (roughly) the first quartile (Q1) and the third quartile (Q3) of the data?

(e) Give the five-numbers summary of the data.

(f) Show a boxplot of the data.

Ans

(a) Mean $\Rightarrow \frac{1}{n}\sum_{i=1}^{n} x_i \Rightarrow \frac{x_i}{n}$

$$\Rightarrow \frac{13+15+16+16+19+20+20+21+22+22+25+25+25+25+30+33+33+35+35+35+35+36+40+45+46+52+70}{27}$$

$$\Rightarrow \frac{809}{27} \Rightarrow 29.96 \approx \boxed{30 = \text{Mean}}$$

Median $\Rightarrow (n+1)/_2$

Sort $\Rightarrow$

13, 15, 16, 16, 19, 20, 20, 21, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46,

$n = 27$

median $\Rightarrow (27+1)/_2$

$\Rightarrow \frac{28}{2} \Rightarrow 14$

14th value $\Rightarrow 25$

$\boxed{\text{Median} = 25}$

(b) Mode & Modality

Mode $\Rightarrow$

Frequencies:

$25 \rightarrow 4$ times

$35 \rightarrow 4$ times

Only these 2 values repeated more

$\boxed{\text{Mode} \Rightarrow 25 \text{ and } 35}$

Since, there are two models, The data is $\boxed{\text{Bimodal}}$ $\Rightarrow$ Modality

(c) Mid range

Q $\Rightarrow$ 13, 15, 16, 16, 19, 20, 20, 21, 22, 25, 25,

$\frac{\text{Min} + \text{max}}{2} \Rightarrow \frac{13 - 70}{2} \Rightarrow \frac{83}{2} \Rightarrow 41.5$

$\boxed{\text{Mid range} \Rightarrow 41.5}$ years.

(d) Q1 & Q3 quantile

Q1 ⇒ 13, 15, 16, 16, 19, 20, 20, 21, 22, 25, 25, 25

$\{Q_1 = 20\}$

Q3 ⇒ 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, [illegible],

$\{Q_3 = 35\}$

(e) Five numbers summary.

Min ⇒ 13

Q1 ⇒ 20

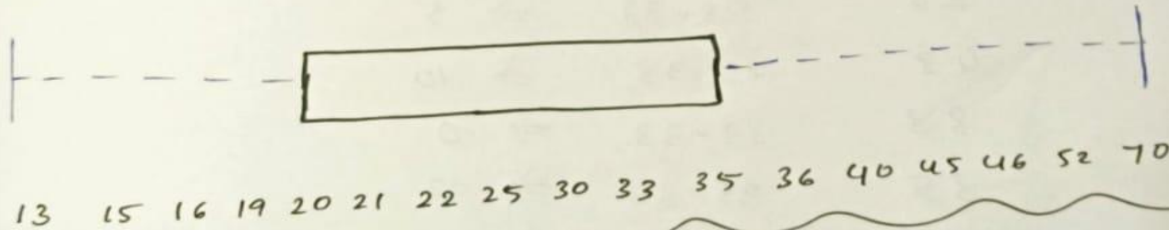
Median ⇒ 25

Q3 ⇒ 35

Max ⇒ 70.

{5-num ⇒ (13, 20, 25, 35, 70)}

(f) Boxplot



2) Given the following measurements for the variable age: 33, 35, 56, 28, 18, 22, 25, 42, 28, 43 the Standardize the variable by the following:

Compute the mean absolute deviation of age.

Ans:

Given ages:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

$n = 10$

mean ⇒ $\frac{\text{Sum of all value}}{n}$ ⇒ $\frac{1}{n}\sum_{i=1}^{n} x_i$

⇒ $\frac{18+22+25+42+28+43+33+35+56+28}{10}$

⇒ $\boxed{\text{Mean} = 33}$

Mean Absolute

x(age)	$\lvert x - \bar{x} \rvert$	
18	18 - 33	⇒ 15
22	22 - 33	⇒ 11
25	25 - 33	⇒ 8
42	42 - 33	⇒ 9
28	28 - 33	⇒ 5
43	43 - 33	⇒ 10
33	33 - 33	⇒ 0
35	35 - 33	⇒ 2
56	56 - 33	⇒ 23
28	28 - 33	⇒ 5

$\boxed{\text{Total} \Rightarrow 88}$

Mean Absolute deviation (MAD)

$MAD = \frac{\sum \lvert x - \bar{x} \rvert}{n} \Rightarrow \frac{88}{10} \Rightarrow 8.8$

MAD = 8.8

(3) Suppose that the data for analysis include the attribute age. The age values for the data tuples ar (in increasing order)
13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.
Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

Ans

Step 1:
Sort ⇒
13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

$$\text{Mean} \Rightarrow \frac{809}{27} \Rightarrow 30.$$

No. of occurences ⇒ 27

Step 2: bin depth ⇒ 3

$$\text{Count of bin} \Rightarrow \frac{\text{occurence}}{\text{bin depth}}$$

$$\Rightarrow \frac{27}{3} \Rightarrow 9$$

Count of bin ⇒ 9

Step 3:

Bin		Original Value		Mean
B1	⇒	13, 15, 16	⇒	$\frac{13+15+16}{3} \Rightarrow 14.67$
B2	⇒	16, 19, 20	⇒	$\frac{16+19+20}{3} \Rightarrow 18.33$
B3	⇒	20, 21, 22	⇒	$\frac{20+21+22}{3} \Rightarrow 21.00$
B4	⇒	22, 25, 25	⇒	$\frac{22+25+25}{3} \Rightarrow 24.00$
B5	⇒	25, 25, 30	⇒	$\frac{25+25+30}{3} \Rightarrow 26.67$
B6	⇒	33, 35, 35	⇒	$\frac{33+33+35}{3} \Rightarrow 33.67$
B7	⇒	35, 35, 35	⇒	$\frac{35+35+35}{3} \Rightarrow 35$
B8	⇒	36, 40, 45, ~~46~~	⇒	$\frac{36+40+45}{3} \Rightarrow 40.33$
B9	⇒	46, 52, 70	⇒	$\frac{46+52+70}{3} \Rightarrow 56$

Step 4: Replace the calculated mean for all 3 values in bin

bin	Smoothed Values
B1	14.67, 14.67, 14.67
B2	18.33, 18.33, 18.33
B3	21.00, 21.00, 21.00
B4	24.00, 24.00, 24.00
B5	26.67, 26.67, 26.67
B6	33.67, 33.67, 33.67
B7	35, 35, 35
B8	40.33, 40.33, 40.33
B9	56, 56, 56

(4) Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000

(a) Min-Max normalization by setting min = 0 and max = 1

(b) Z-score normalization

Sol:

(a) Min-Max Normalization (Min=0 & max=

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Min Value ⇒ 200

Max Value ⇒ 1000

$$x' \Rightarrow \frac{x - 200}{1000 - 200} \Rightarrow \frac{x - 200}{800}$$

$$x' = \frac{x - 200}{}$$

Original Value (x)	Calculation	Normalized Value (x')
200	$(200-200)/800$	0.000
300	$(300-200)/800 \Rightarrow \frac{100}{800}$	0.125
400	$\frac{(400-200)}{800} \Rightarrow \frac{200}{800}$	0.250
600	$\frac{(600-200)}{800} \Rightarrow \frac{400}{800}$	0.500
1000	$\frac{(1000-200)}{800} \Rightarrow \frac{800}{800}$	1.000

(b) Z-Score Normalization

$$z = \frac{x - \mu}{\sigma}$$

Step 1: Mean

$$\mu \Rightarrow \frac{200+300+400+600+1000}{5}$$

$$\Rightarrow \frac{2500}{5} \Rightarrow \boxed{500 = \mu}$$

Step 2: Standard deviation

x	$(x-\mu)$	$(x-\mu)^2$
200	$(200-500) \Rightarrow (-300)$	90,000
300	$(300-500) \Rightarrow (-200)$	40,000
400	$(400-500) \Rightarrow (-100)$	10,000
600	$(600-500) \Rightarrow (100)$	10,000
1000	$(1000-500) \Rightarrow (500)$	250,000

$$\boxed{\Sigma(x-\mu)^2 \Rightarrow 4,00,000}$$

population Variance

$$\sigma^2 = \frac{400,000}{5} \Rightarrow 80,000$$

Standard deviation:

$$\sigma = \sqrt{80,000} \Rightarrow 282.84$$

Step 3 : compute z - Scores

original Value	Calculation $\left(\frac{x-\mu}{\sigma}\right)$	z - score
200	$(-300)/282.84$	-1.06
300	$(-200)/282.84$	-0.71
400	$(-100)/282.84$	-0.35
600	$(100)/282.84$	0.35
~~800~~	~~300/(282.84)~~	~~1.06~~
1000	$500/282.84$	1.77

$\therefore$ z scores are

$-1.06, -0.71, -0.35, 0.35, 1.77$

(5) Suppose a group of 12 Sales price records has been Sorted as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into three bins by each of the following methods.

(a) equal - frequency partioning

(b) equal - width partioning

(c) clustering.

5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215

Sol:

Total record ⇒ 12

Bins ⇒ 3

(a) Equal - frequency partioning.

No. of records of bins ⇒ $\frac{12}{3}$ ⇒ $\boxed{4}$

bins	Values
B1	5, 10, 11, 13
B2	15, 35, 50, 55
B2	72, 92, 204, 215

(b) Equal - width partioning.

$$width \Rightarrow \frac{Max - Min}{No.\ of\ bins}$$

$$\Rightarrow \frac{215 - 5}{3} \Rightarrow \frac{210}{3} \Rightarrow 70$$

$\boxed{width = 70}$

The intervals:

- Bin 1 ⇒ 1 - 74
- Bin 2 ⇒ 75 - 144
- Bin 3 ⇒ 145 - 215

Now, place the values into the intervals

Bin	Interval	Values
Bin 1	1 - 74	5, 10, 11, 13, 15, 35, 50, 55, 72
Bin 2	75 - 144	92
Bin 3	145 - 215	204, 215

(c) Clustering

Cluster (Bin)	Value
cluster 1	5, 10, 11, 13, 15
cluster 2	35, 50, 55, 72, 92
cluster 3	20.5, 215

(6) Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result

age	23	23	27	27	39	11	47	49
% fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2

age	52	54	54	56	57	58	58	60
% fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2

(a) Calculate the mean, median, & standard deviation of age and % fat

(b) Draw the boxplots for age & % fat

(c) Draw a scatter plot and a q-q plot based on these two variables

(d) Normalize the two variables based on z-score normalization

(e) Calculate the correlation coefficient (person's product moment coefficient) Are these two variables positively or negatively correlated?

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sol

(a) Mean, Median, Standard deviation

age		% fat
23	→	9.5
23	→	26.5
27	→	7.8
27	→	17.8
39	→	31.4
41	→	25.9
47	→	27.4
49	→	27.2
50	→	31.2
52	→	34.6
54	→	42.5
54	→	28.8
56	→	33.4
57	→	30.2
58	→	34.1
58	→	32.9
60	→	41.2
61	→	35.7

∴ no. of observations

$n \Rightarrow 18$

$$\text{Mean (age)} = \frac{\Sigma x}{n} = \frac{836}{18}$$

$$\boxed{\text{mean (age)} \Rightarrow 46.44}$$

$$\text{mean (\% fat)} = \frac{\Sigma x}{n} \Rightarrow \frac{518.1}{18}$$

$$\boxed{\text{Mean (\% fat)} \Rightarrow 28.78}$$

$$\text{Median} \Rightarrow \frac{50+52}{2} \Rightarrow 51$$

$$\boxed{\text{Median} \Rightarrow 51}$$

(age) Standard Deviation ⇒

$$S = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$$

$$S = \frac{\sqrt{(46.44)^2}}{18-1}$$

$$\boxed{S = 13.22}$$

$$\text{Std (\% fat)} \Rightarrow S = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$$

$$\boxed{S = 9.25}$$

Variable	Mean	Median	Standard Deviation
Age	46.44	51	13.22
% fat	28.78	30.70	9.25

(b) Box plot

Age (five number Summary)

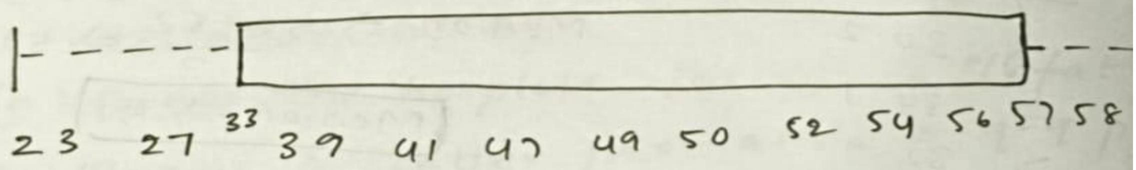
Min ⇒ 23
Q_1 ⇒ 33
Median ⇒ 51
Q3 ⇒ 57
Max ⇒ 61

Q_1 ⇒ 33 ⇒
Median ⇒ 5
Q2 ⇒ 57 ⇒

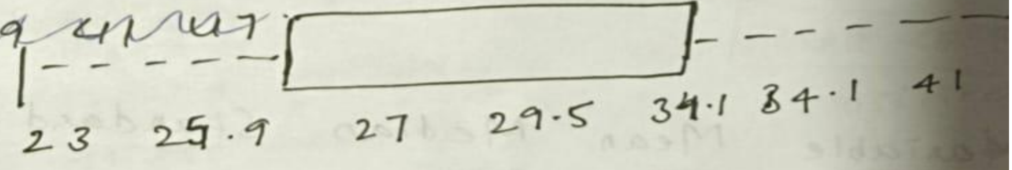
%Fat (5-num Summary)

Min ⇒ 7.8
Q_1 ⇒ 26.5
Median = 30.7
Q3 = 34.1
Max = 42.5

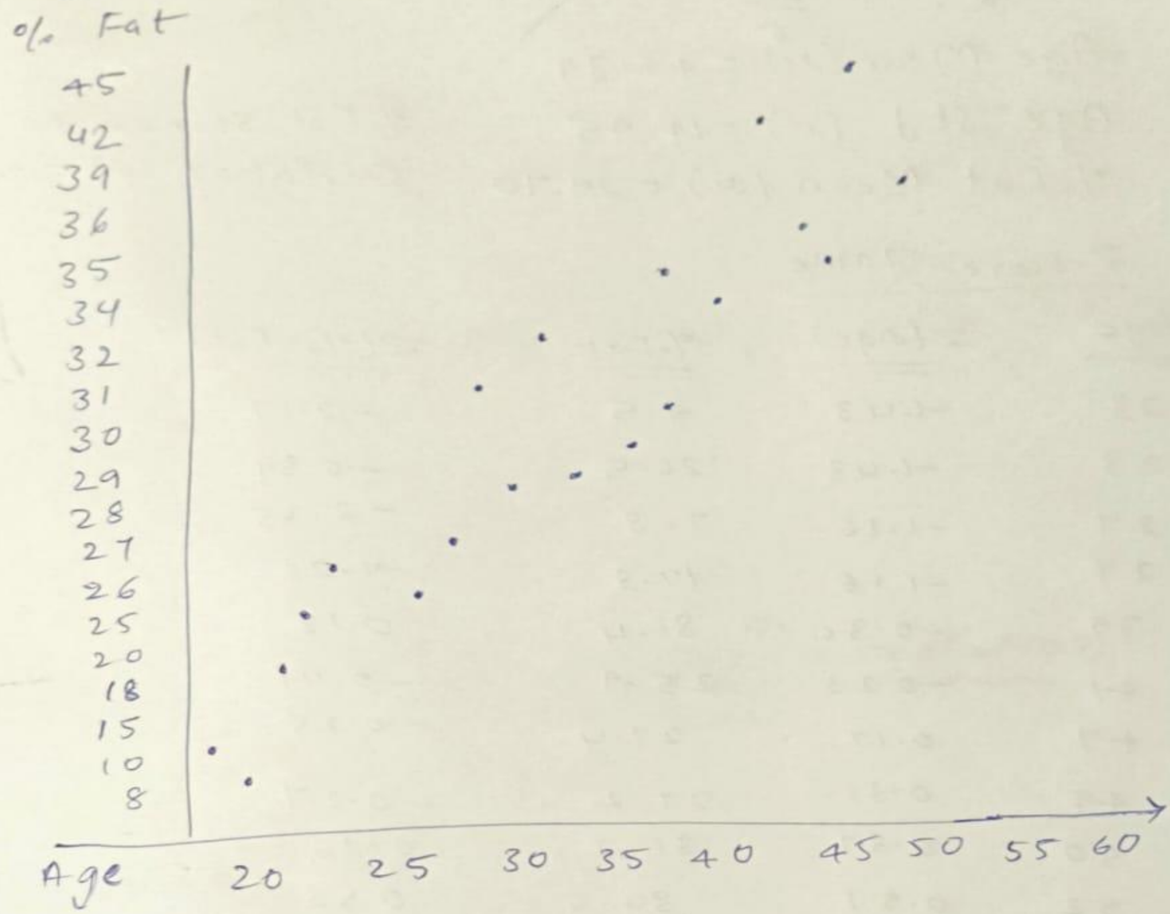
Age



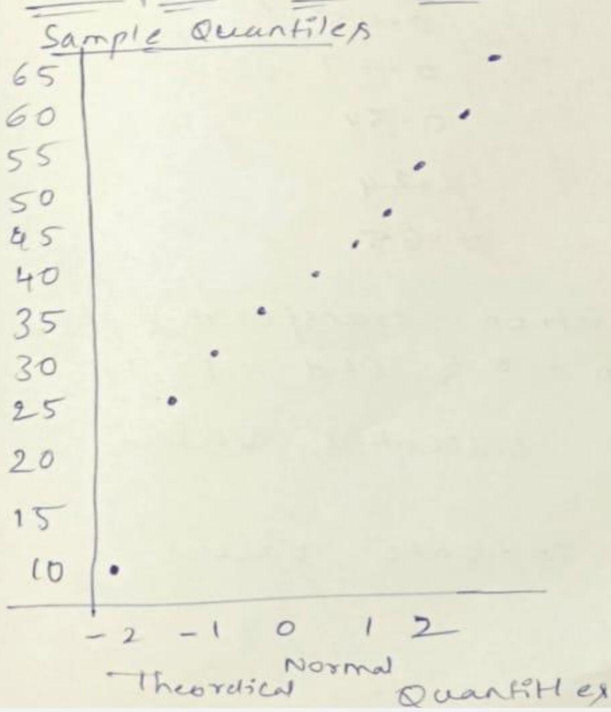
% fat



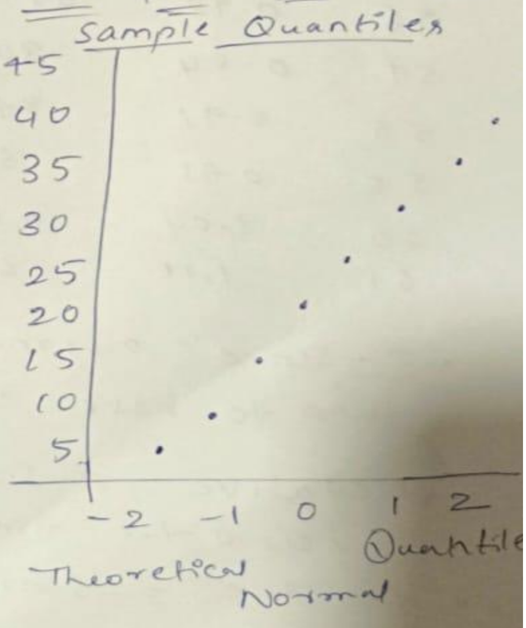
(c) Scatter plot & q-q. plot



Q-Q plot for Age



Q-Q plot for % Fat



ⓓ Z-Score Normalization

$$Z = \frac{x - \mu}{\sigma}$$

Age Mean (μ) = 44.39

Age Std (σ) = 14.95

% Fat Mean (μ) = 29.70

% Fat Standard Deviation (σ) = 9.

Z-Score Table

Age	Z(Age)	% Fat	Z(% Fat)
23	-1.43	9.5	-2.17
23	-1.43	26.5	-0.34
27	-1.16	7.8	-2.35
27	-1.16	17.8	-1.28
39	-0.36	31.4	0.18
41	-0.23	25.9	-0.41
47	0.17	27.4	-0.25
49	0.31	27.2	-0.27
50	0.37	31.2	0.16
52	0.51	34.6	0.53
54	0.64	28.8	-0.10
56	0.78	33.4	0.40
57	0.84	30.2	0.05
58	0.91	34.1	0.47
58	0.91	32.9	0.34
60	1.04	41.2	1.24
61	1.11	35.7	0.65

- Z-Score normalization transforms t data to have mean = 0 & Std = 1
- Negative Z-Scores indicate values below the mean.
- Positive Z-Scores indicate values above the mean.

(e) Calculate the pearson correlation coefficient

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)}}$$

- number of observation $(n) = 18$
- person corelation coefficient $(r) \approx 0.82$

Interporetation

Since:

- $r = 0.82$
- $r > 0$

-there is a strong positive correlation between Age and % Fat.

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation